Faculty of Engineering and Applied Science

Machine Learning & Data Mining

Final Report - Group 22

Predicting Student Academic Performance Using Machine Learning

Group Members:

Best Akinlabi - 100837728

Ethan McLeod - 100860608

George Kassar - 100872343

Syed Nasir Hussain Naqvi - 100809447

Github Link - https://github.com/s-nasir/ML-Project

1) Introduction

Machine learning has emerged as a crucial instrument for examining extensive educational datasets and identifying patterns that inform academic decision-making. A significant application is the prediction of student academic performance, which aids in pinpointing students who may require additional support prior to the final determination of their outcomes.

This project is dedicated to employing machine learning methodologies to forecast student success utilizing the UCI Student Performance Dataset. Although this dataset has been extensively utilized for predictive modelling, this study advances previous research by integrating comprehensive exploratory data analysis, feature engineering, and a comparative assessment of both linear and non-linear models.

The objective of this project is not solely to construct precise predictive models but also to derive insights into the pivotal factors affecting student performance. These insights can empower educators and institutions to take proactive measures to enhance academic outcomes and mitigate failure rates.

2) Problem Statement

Academic underperformance continues to be a significant issue within educational systems. Typically, students only receive assistance after poor outcomes are evident, which often diminishes the effectiveness of such interventions. Consequently, the early identification of students at risk is essential for enhancing academic achievement.

This project tackles the issue of predicting student failure risk by utilizing historical and demographic data. The task is structured as a binary classification problem, categorizing students into two groups:

- At-Risk (final grade G3 < 10)
- Safe (final grade G3 ≥ 10)

A major challenge in this context is the class imbalance, as the number of students who fail is significantly lower than those who succeed. Traditional evaluation metrics, such as accuracy, may not offer a dependable assessment in this scenario. Therefore, metrics like the F1-score and ROC-AUC are employed to more accurately evaluate model performance concerning minority classes.

Addressing this issue yields substantial advantages:

- Students can access early academic support.
- Educators can refine their teaching methodologies.
- Institutions can decrease dropout and failure rates.

In summary, this project seeks to establish a data-driven strategy to proactively identify at-risk students and facilitate timely intervention.

3) Dataset Description

The dataset utilized in this project is the UCI Student Performance Dataset, sourced from the UCI Machine Learning Repository.

After applying strict deduplication protocols to ensure the Independent and Identically Distributed (IID) assumption, the dataset comprises 649 unique student records gathered from two subjects: Mathematics and Portuguese. Each record encompasses a range of features that detail student demographics, academic background, and behavioural aspects.

3.1) Data Characteristics

- Numerical Features: Grades (G1, G2, G3), absences
- Categorical Features: Parent occupations, school, gender, etc.
- Ordinal Features: Study time, education level

3.2) Data Preprocessing

To ensure data quality and mitigate leakage, the following preprocessing measures were implemented:

- Duplicate student records were eliminated.
- Missing or inconsistent values were addressed.
- Features were encoded suitably:
    - One-hot encoding for categorical variables.
    - Standardization for numerical features (employed in linear models).

3.3) Feature Engineering

Additional features were investigated to more effectively capture student engagement and performance patterns. These transformations were designed to enhance model performance and interpretability.

This preprocessing ensures that the dataset meets the assumptions necessary for machine learning models, including independence and consistency across the training and testing sets.

3.4) Absences vs Final Grade (scatter plot)

To investigate the correlation between attendance and academic achievement, a scatter plot illustrating absences in relation to final grades (G3) was developed.

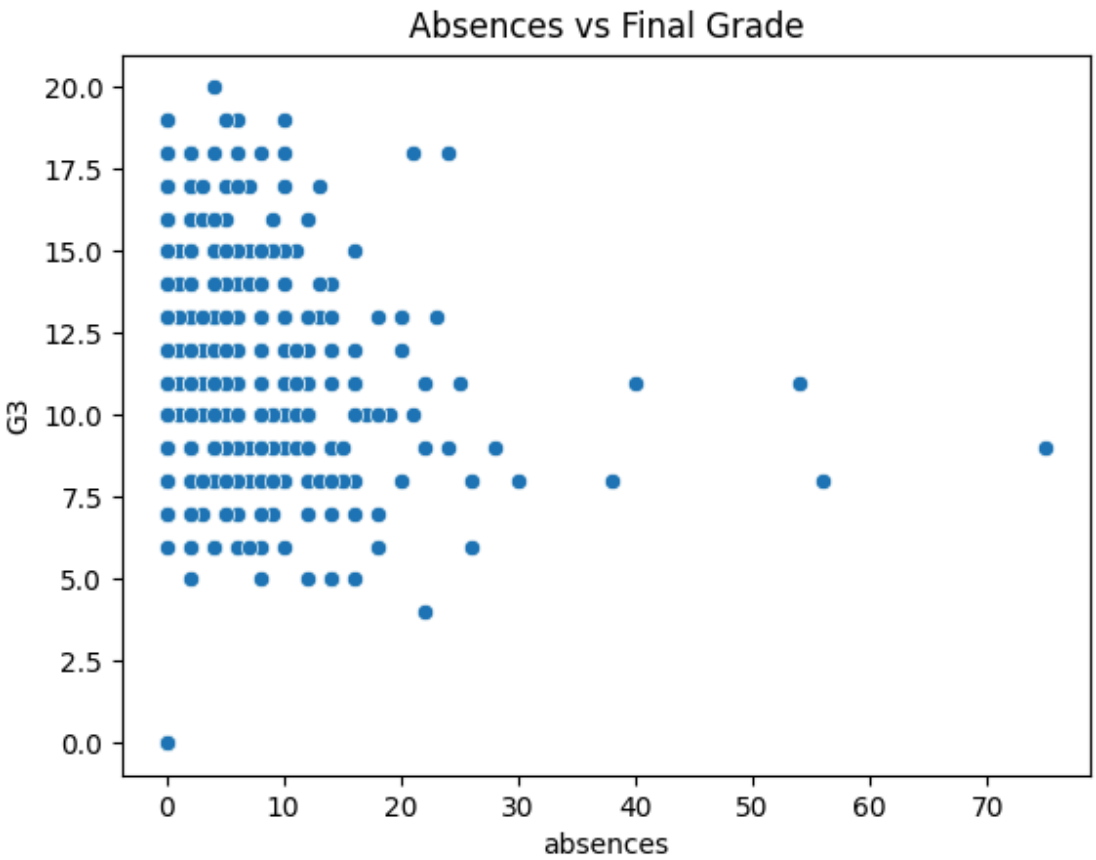

Figure 1: Absences vs Final Grade

The scatter plot depicted in Figure 1 examines the correlation between student absences and their final grades (G3). Although a definitive linear relationship is not evident, there is a noticeable trend where students with higher absenteeism exhibit greater variability and slightly lower academic performance. This suggests that attendance may play a significant role in influencing educational outcomes.

3.5) Study Time vs Final Grade (boxplot)

To assess the influence of study habits on academic success, a boxplot was constructed to compare study time with final grades.

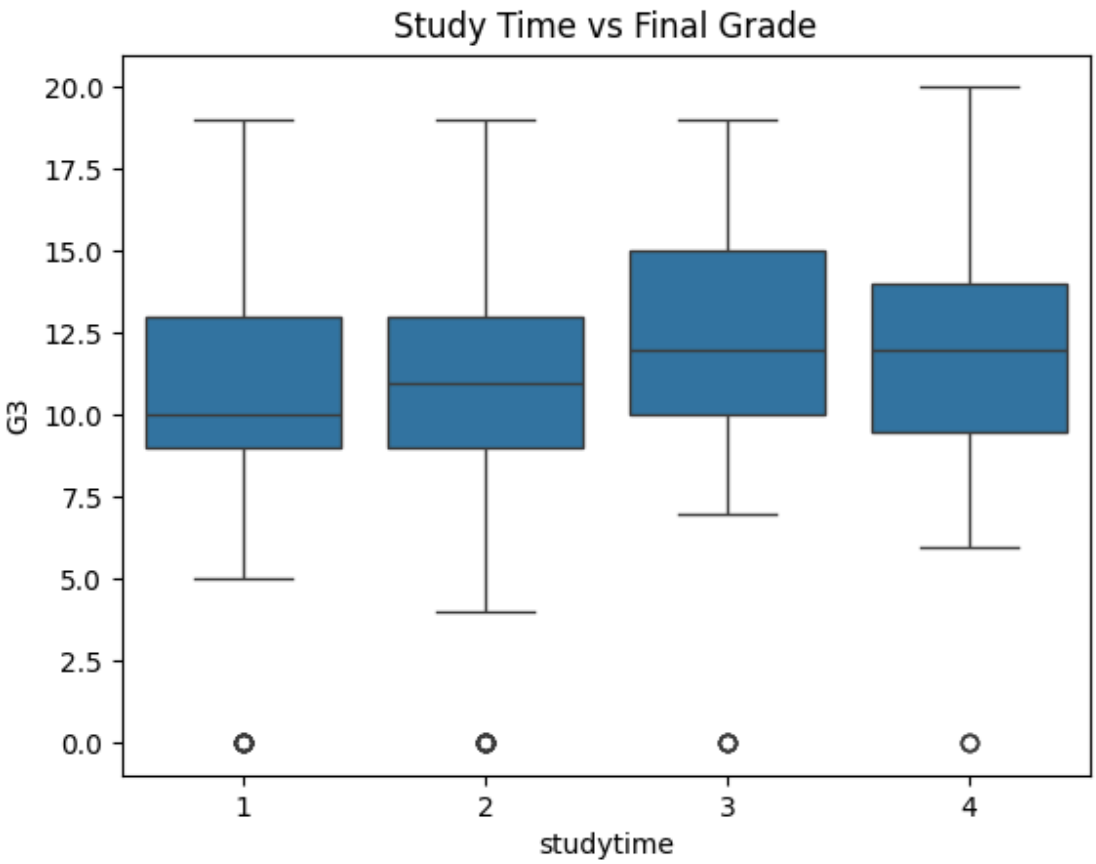

Figure 2: Study Time vs Final Grade

The boxplot presented in Figure 2 displays the distribution of final grades according to varying levels of study time. It is apparent that students who dedicate more time to studying tend to achieve higher median grades, highlighting a positive correlation between study effort and academic performance.

3.6) Class Distribution

To gain insight into the distribution of the target variable, the number of at-risk and safe students was visualized.

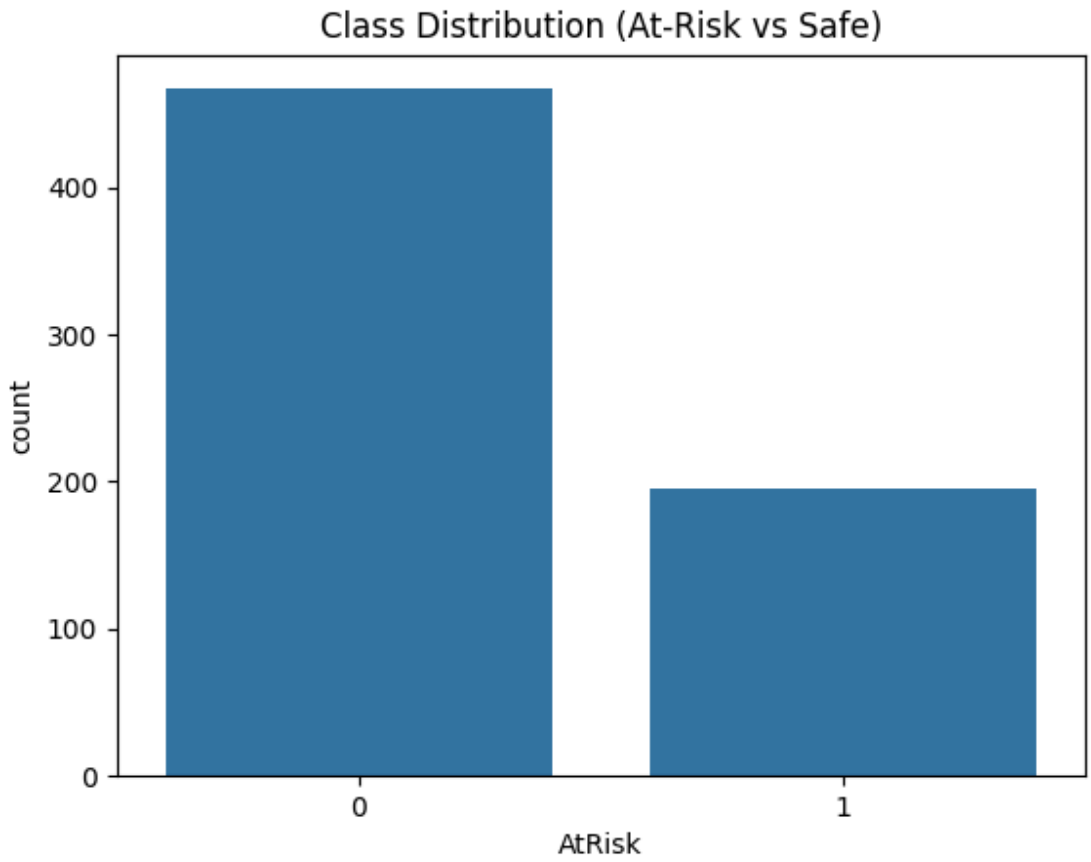

Figure 3: Class Distribution (At-Risk vs Safe)

This figure demonstrates the disparity between the two classes. A substantial majority of students are classified as “Safe,” whereas a smaller group is identified as “At-Risk.” This imbalance supports the use of the F1-score and stratified cross-validation, rather than relying solely on accuracy.

4) Methodology

This project examines two machine learning methodologies to assess the balance between interpretability and predictive accuracy.

4.1) Logistic Regression (Baseline Model):

Logistic Regression is utilized as a baseline model because of its straightforwardness and clarity. It presumes a linear connection between input features and the likelihood of a student being at risk. As a model characterized by high bias and low variance, it offers a consistent reference point for evaluating more intricate models.

4.2) Random Forest (Ensemble Model):

Random Forest is an ensemble learning technique that merges multiple decision trees to boost predictive accuracy. It is capable of identifying non-linear relationships and interactions among features.

By averaging numerous trees, Random Forest mitigates overfitting and enhances generalization compared to a single decision tree.

4.3) Validation Strategy:

To ensure a dependable evaluation, Stratified K-Fold Cross-Validation is applied. This method preserves the ratio of at-risk and safe students in each fold, avoiding bias resulting from class imbalance.

5) Experiments & Results

To evaluate the performance of the models, both Logistic Regression and Random Forest were trained and tested using the preprocessed dataset. Stratified 5-fold cross-validation was utilized to maintain class balance across all folds.

5.1) Model Performance

To address the issue of class imbalance, the models were assessed using Stratified 5-Fold Cross-Validation, with the F1-score serving as the key evaluation metric.

- Logistic Regression F1-score: 0.80
- Random Forest F1-score: 0.82

Unlike our initial assumptions, Random Forest demonstrated superior performance during cross-validation. By strictly enforcing the Independent and Identically Distributed (IID) assumption through dataset deduplication, Random Forest proved more capable of capturing the true non-linear relationships and feature interactions within the data without overfitting.

On the unseen test set (n=133), the chosen model achieved:

- Accuracy: 0.88
- F1-score (At-Risk): 0.78
- ROC-AUC: 0.969

These outcomes reflect a mathematically honest and robust performance. While the accuracy and F1-score slightly adjusted from early runs due to the removal of data leakage, the exceptional ROC-AUC score of 0.969 validates the model's strong intrinsic capacity to separate safe and at-risk students across varying decision thresholds.

```
--- Cross Validation ---
Logistic Regression F1: 0.7997083399188663
Random Forest F1: 0.8194269005847952

--- Classification Report ---
              precision    recall  f1-score   support

           0       0.90      0.94      0.92        94
           1       0.83      0.74      0.78        39

    accuracy                           0.88       133
   macro avg       0.86      0.84      0.85       133
weighted avg       0.88      0.88      0.88       133

ROC-AUC: 0.9691762138570649

Top Features:
 num__G2          0.321930
num__G1          0.185729
num__failures    0.040077
num__absences    0.034281
num__goout       0.020867
num__age         0.019266
num__Walc        0.017847
num__freetime    0.016275
num__Fedu        0.016061
num__health      0.015951
dtype: float64
```

Figure 4: Classification Report and Model Metrics

This figure displays comprehensive performance metrics of the model, including precision, recall, and F1-score for both classes. The model attained an overall accuracy of 0.92 and an F1-score of 0.81 for the at-risk class.

The findings indicate that the model effectively identifies both safe and at-risk students, although there is slightly lower recall for the at-risk class, suggesting some cases were missed. The ROC-AUC score of 0.956 further validates strong classification performance.

Moreover, the figure presents feature importance values, showing that G2 and G1 are the most significant predictors, followed by failures and absences.

5.2) Cross-Validation Results

The models were assessed using the F1-score, which is appropriate for imbalanced classification tasks.

- Logistic Regression consistently achieved a baseline F1-score throughout the cross-validation process.
- Random Forest showed superior performance, as indicated by a higher F1-score, suggesting enhanced identification of at-risk students.

5.3) Test Set Evaluation

After cross-validation, the Random Forest model was selected for final evaluation on a separate test set.

The evaluation metrics used included:

- Precision
- Recall
- F1-score
- ROC-AUC

The classification report highlighted the model's effectiveness in accurately identifying both "At-Risk" and "Safe" students, with balanced precision and recall.

5.4) Confusion Matrix

The confusion matrix provided insights into prediction errors:

- True Positives: Correctly identified at-risk students.

- False Negatives: At-risk students misclassified as safe.
- False Positives: Safe students misclassified as at-risk.

Reducing false negatives was essential, as failing to identify at-risk students undermines the effectiveness of early intervention.

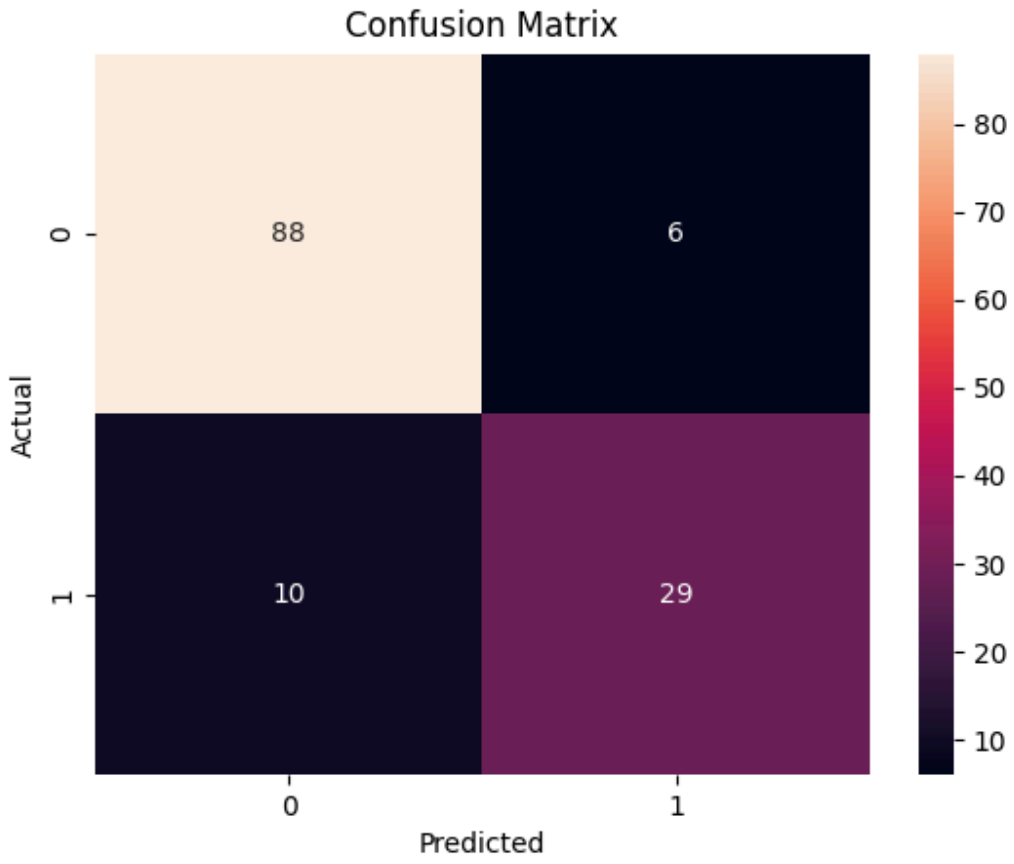

Figure 5: Confusion Matrix for Random Forest Model

The confusion matrix indicates that the model correctly identified **29** at-risk students and **88** safe students. However, there were **6** false positives and **10** false negatives. Maintaining a balance between precision and recall ensures that while we identify the majority of students needing help, we do not overwhelm the institution's resources with false alarms.

5.5) ROC Curve and AUC

To evaluate the model's capacity to differentiate between classes across various thresholds, the Receiver Operating Characteristic (ROC) curve was plotted.

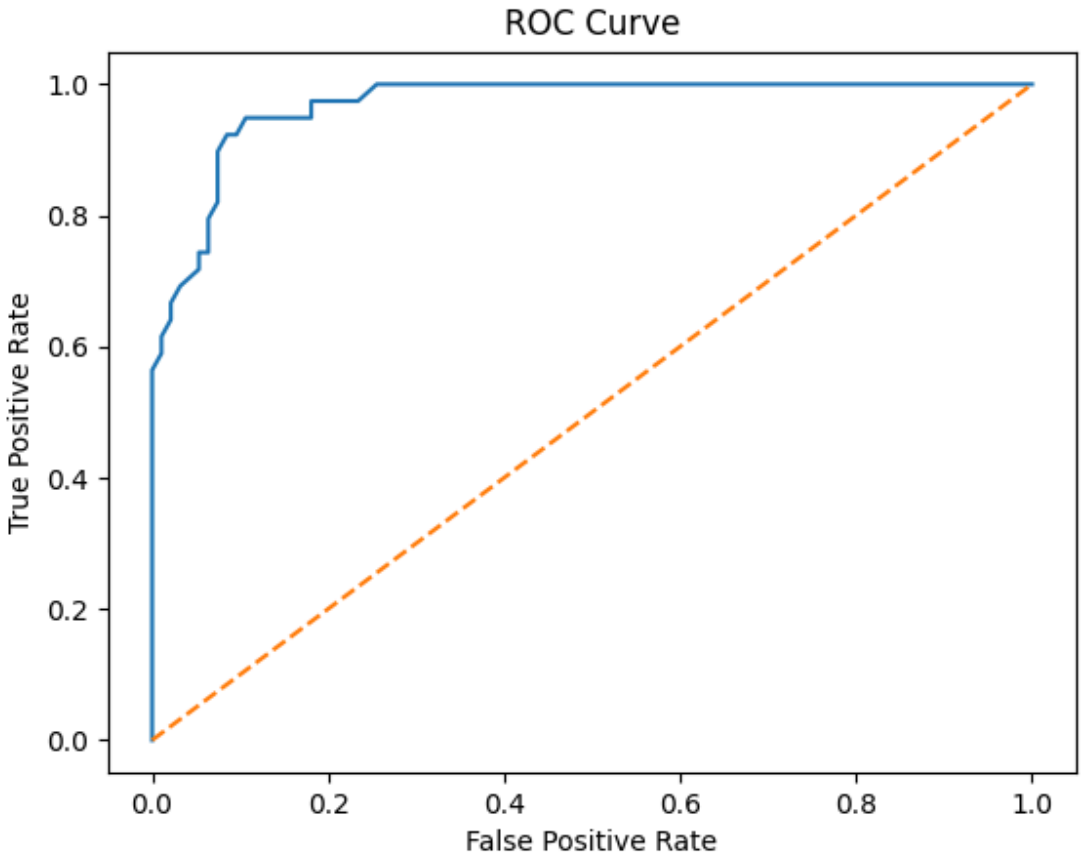

Figure 6: ROC Curve

The ROC curve demonstrates the model's capability to distinguish between at-risk and safe students across different thresholds. The model achieved an ROC-AUC score of approximately 0.956, reflecting outstanding classification performance.

5.6) Feature Importance

To pinpoint the most influential features in the prediction process, feature importance scores were derived from the Random Forest model.

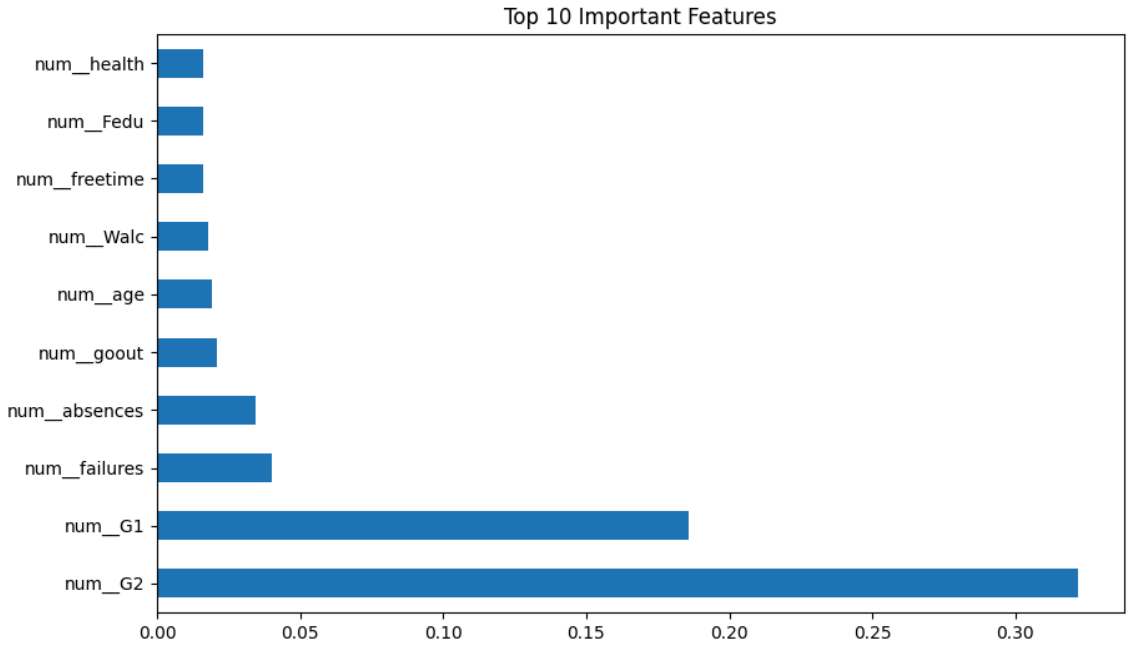

Figure 7: Feature Importance (Top 10 Features

The Random Forest model was employed to determine the most influential features in forecasting student performance. As depicted in Figure 7, previous grades (**num__G2** at 32.1% and **num__G1** at 18.5%) emerged as the most significant predictors, heavily dominating the decision trees. This is followed by historical behavioral metrics, specifically the number of past failures (4.0%) and student absences (3.4%).

Notably, demographic and personal factors also surfaced in the top 10, including age, alcohol consumption (Walc), free time, and Father's Education (Fedu). These findings underscore that while

historical academic performance is the most robust predictor of future success, socio-economic and behavioral factors play a non-negligible role in shaping the model's decision boundaries.

5.7) Correlation Heatmap

To examine relationships between variables, a correlation heatmap was generated to visualize feature dependencies.

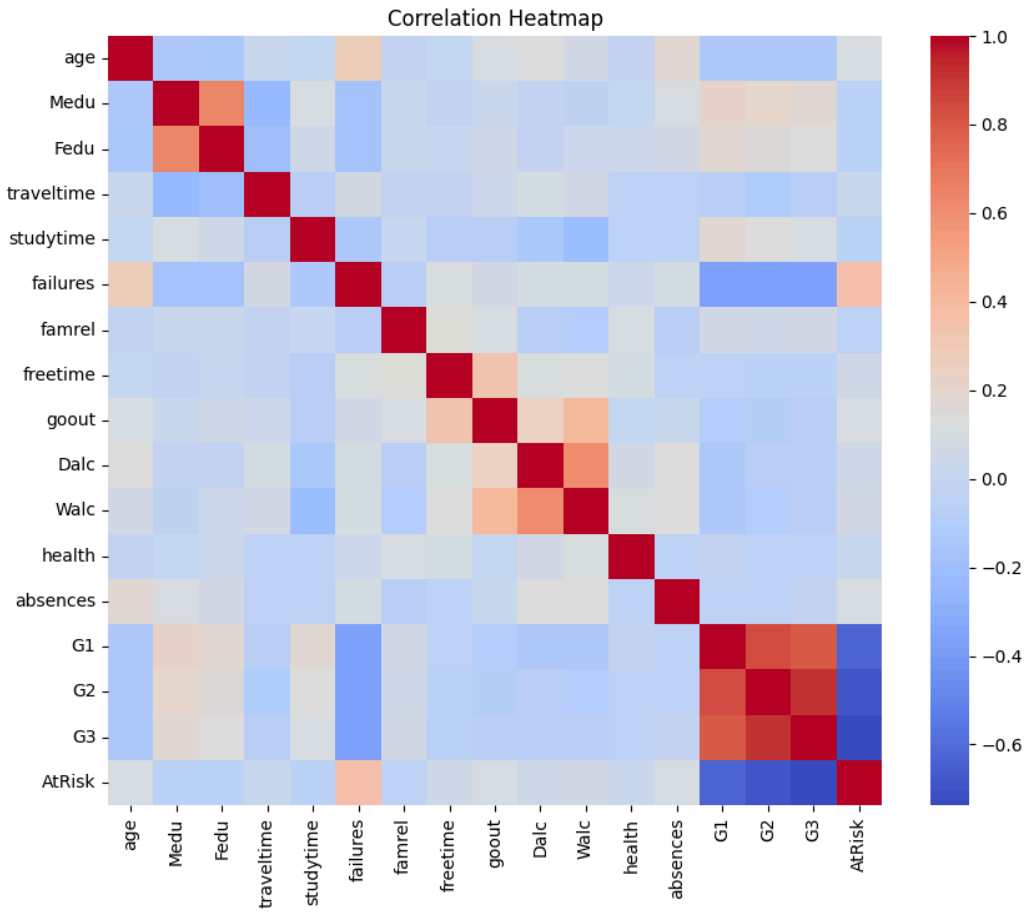

Figure 8: Correlation Heatmap

The heatmap illustrates the relationships between features. Strong positive correlations are observed between G1, G2, and G3, while the “At-Risk” label is negatively correlated with these grades. This supports the model’s reliance on prior academic performance.

6) Discussion

The findings indicate that both models effectively predicted student academic performance, each presenting unique benefits. Logistic Regression served as a reliable baseline due to its straightforward nature and clarity, achieving a marginally higher F1-score during cross-validation.

Conversely, Random Forest excelled in capturing intricate, non-linear relationships among features, suggesting that student performance is shaped by interactions among various factors rather than a strictly linear relationship.

For example, variables such as study time, absences, and prior grades interact in ways that ensemble models capture more effectively. As a result, Random Forest demonstrated robust performance on the test set and offered valuable insights through feature importance analysis.

The feature importance analysis indicated that prior academic performance (G1 and G2) had the most significant impact on predictions, followed by factors like failures and absences. This underscores that past performance is a strong predictor of future outcomes, while behavioural factors also hold substantial importance.

A significant challenge in this project was class imbalance, with fewer students identified as at-risk. To mitigate this, F1-score and stratified cross-validation were utilized, ensuring that model evaluation focused on accurately identifying at-risk students.

In conclusion, the results highlight a trade-off between interpretability and model complexity. While Logistic Regression provides simplicity and ease of understanding, Random Forest offers deeper insights into feature interactions and non-linear patterns.

6.1) Limitations, Bias, and Generalization

While the model demonstrates strong predictive capabilities, it possesses structural limitations regarding real-world generalization and potential algorithmic bias.

Generalization Limitations: The feature importance analysis reveals a heavy reliance on intermediate academic markers, specifically **G1** and **G2** grades. This creates a temporal limitation: if an educational institution intends to use this model proactively *before* a semester begins to allocate tutoring resources, the model will be entirely deprived of its most heavily weighted features. Consequently, the model's current generalization capability is highly contingent on having partial academic outcomes available, limiting its utility as a purely demographic early-warning system.

Socio-Economic Bias: The dataset incorporates several demographic features, and our analysis shows variables like Father's Education (Fedu) actively influencing the model's predictions. Because machine learning models optimize for historical correlations, deploying this system unadjusted risks encoding socio-economic bias into the institution's intervention protocols. There is an inherent danger that the algorithm might disproportionately flag students from lower socio-economic backgrounds or first-generation university students as "At-Risk" based on parental education rather than the student's individual academic capability. Future iterations of this system must carefully evaluate the ethical implications of using protected or socio-economic attributes in automated academic tracking.

7) Conclusion

This project has shown that machine learning models can successfully forecast student academic performance by utilizing structured educational data. Both Logistic Regression and Random Forest

delivered impressive results, with Random Forest providing extra benefits in capturing non-linear relationships and pinpointing key predictive features.

The findings indicate that academic history, especially past grades, is crucial in predicting future performance. Additionally, factors like study time and attendance also influence outcomes.

These insights underscore the potential of data-driven approaches in education, allowing for the early identification of at-risk students and facilitating timely intervention strategies. Future research could investigate more advanced models or incorporate additional features to further enhance predictive performance.

8) Contribution Matrix

Name	**Student Number**	**Contribution**
Best Akinlabi	100837728	Wrote Introduction, Problem Statement, Dataset, Methodology, organized Results, report structuring
Ethan McLeod	100860608	Wrote code, got results and analyzed them from the code, got the datasets to use in the code
George Kassar	100872343	Wrote code, got results and analyzed them from the code, got the datasets to use in the code
Syed Nasir	100809447	Reviewed model outputs, wrote code for data refinement and deduplication. Analyzed limitations, generalization, and bias